

Probabilistic Machine Learning

Lecture 04: Naive Bayes and the Generative Route

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Speaker notes

The lecture's place in the arc: after Lecture 03's two shortcuts (keep the peak; model only $p(y \mid x)$), the joint returns — and immediately forces the third shortcut, "assume it apart". Three modules: the generative route and its explosion; naive Bayes, read entirely through earlier lectures (fitting = MLE of a categorical, smoothing = the Beta posterior mean, double counting = over-confidence); and the explicit contrast with logistic regression — the same linear log-odds, fitted two ways.

2026-09-26 (course plan N2): conditional independence properly and the graphical-models introduction moved to the start of Lecture 05. This lecture defines conditional independence only as far as naive Bayes needs it. Fraud is the running example throughout; the shared Data Science units (bayes-decision-making, bayes-learning, naive-bayes) are replaced in PML by three PML units and stay unchanged for Data Science.



Where We Left Off

Lecture 03 took two shortcuts and named what each drops:

Shortcut	Instead of the ideal	What is dropped
Keep the peak	the most probable parameter (MLE, MAP)	the spread: how sure we are
Only what you need	model $p(y \mid x)$ directly — logistic regression	$p(x)$: every question about the inputs

Today the inputs come back. The **generative route** models how the data arises, class by class, and converts with Bayes:

$$p(x, y) = p(y) \cdot p(x \mid y)$$

THE CATCH

$p(x \mid y)$ is a joint over every attribute — and it explodes. Paying for it is **shortcut 3: assume it apart.**

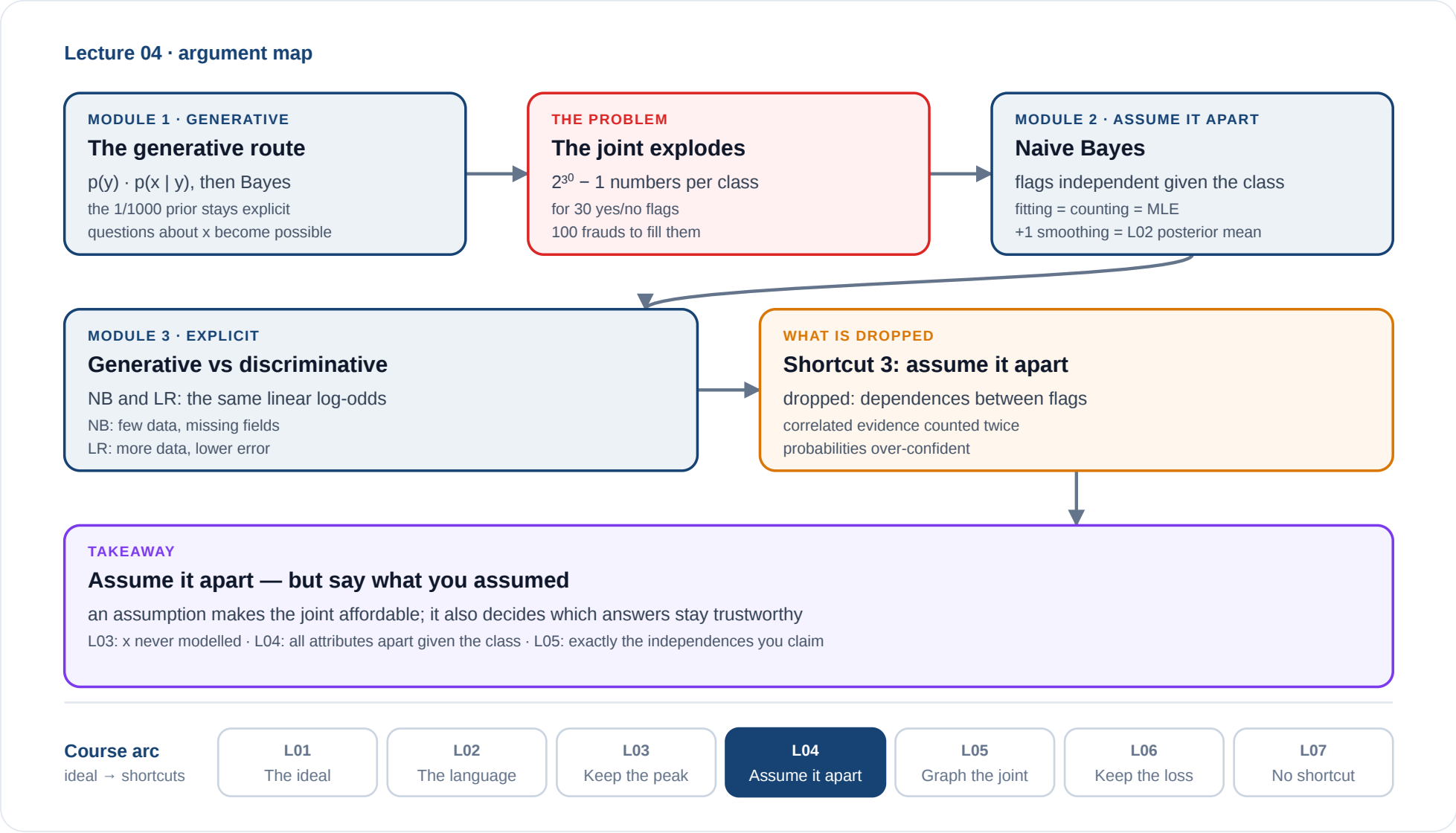
Speaker notes

Return to the shortcut table of Lecture 01, as promised there. Two rows are done; today's row is "assume it apart". Say the plan out loud: Module 1 takes the generative route and hits the explosion; Module 2 pays with the naive independence assumption and reads every step of naive Bayes as something the students already know (MLE, the Beta posterior of L02); Module 3 puts naive Bayes and logistic regression side by side and shows they are the same linear model, fitted two ways.

The fraud example of Lectures 01–02 returns as the running example: a base rate of 1 in 1,000 and a handful of yes/no red flags.



Argument Map



The lecture on one page. Top row: the generative route, the explosion it causes, and naive Bayes as the payment. Middle row: the explicit contrast with logistic regression and the shortcut named with what it drops. Bottom row: the takeaway. Point at the strip: Lecture 04 is "assume it apart".



The Generative Route

Guiding Question: Logistic regression never asked what a fraudulent transaction *looks like*. What would it take to model that — and why does nobody do it in full?

Two Routes to the Same Posterior

A labelled table of transactions supports **both** routes to $P(\text{fraud} \mid x)$, where x is the vector of red flags:

DISCRIMINATIVE — LECTURE 03

Model $p(y \mid x)$ directly

Logistic regression: a score for each x .
Nothing is said about how x arises.

GENERATIVE — TODAY

Model $p(y)$ and $p(x \mid y)$

How does a *fraudulent* transaction look, and how an honest one? Then convert with Bayes.

$$P(y \mid x) = \frac{p(x \mid y) \cdot P(y)}{\sum_{y'} p(x \mid y') P(y')}$$

Speaker notes

Math first on the second pass: the students know Bayes' theorem from L02; the only new thing is that the "hypothesis" is now a class and the "evidence" a whole attribute vector. The colour slots are those of L02 (posterior, likelihood, prior, normaliser).

Correct a misconception before it forms: the generative route is not forced on us by the data. A labelled table estimates $p(y | x)$ just as directly — Lecture 03 did exactly that. The generative route is a *choice*: model each class separately, then convert.



Why Model How Fraud Looks?

Logistic regression already gives $P(\text{fraud} \mid x)$. What can a model of $p(x \mid y)$ do that it cannot?

Speaker notes

Let the room answer before the click; students with an ML background usually name "missing values" and sometimes "the prior". The novelty point is the one they rarely name, and it is exactly what "only what you need" dropped in Lecture 03.

Each of the three items returns in Module 3 as a concrete comparison. Here they only motivate paying the price on the next slide.



Why Model How Fraud Looks?

Logistic regression already gives $P(\text{fraud} \mid x)$. What can a model of $p(x \mid y)$ do that it cannot?

- ✓ **Keep the prior explicit.** The 1 in 1,000 of Lecture 02 is a separate factor: a new market with 1 in 200 changes $P(y)$, not the whole model.
- ✓ **Handle a missing field.** An attribute nobody observed can be summed out of $p(x \mid y)$.
- ✓ **Notice the strange.** A transaction with tiny $p(x)$ under *both* classes is unlike anything seen — a score alone cannot say so.

The price: $p(x \mid y)$ is a **joint distribution over all attributes**, one per class.

The Joint Explodes

30

yes/no red flags per transaction



$2^{30} - 1$

≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone



100

frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Speaker notes

The numbers: $2^{30} - 1 = 1,073,741,823$ free parameters per class (the cells of a joint over 30 binary variables, minus one because they sum to 1). With 100 frauds, at most 100 cells are non-empty. Connect to Lecture 03's three-heads-in-three-flips: MLE on too little data gives extreme answers; here it gives zeros almost everywhere.

The way out is not more data — no bank has a billion frauds. It is an assumption that makes the table small, which is Module 2.



Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

Head arithmetic on purpose. If someone answers 16 or 17, the "minus one" (normalisation) is the point to make; it is the same count as the $2^n - 1$ above. Keep the 15 on the board: Module 2 reduces it to 7.



Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x | y)$ — and converts with Bayes. It keeps the prior explicit and can answer questions about x itself, but it must pay for a **joint over all attributes**, which explodes.

Naive Bayes: Assume It Apart

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

Definition. A and B are **conditionally independent given C** , written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \iff P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A .**

Among frauds only: does a new device make a foreign IP more likely?

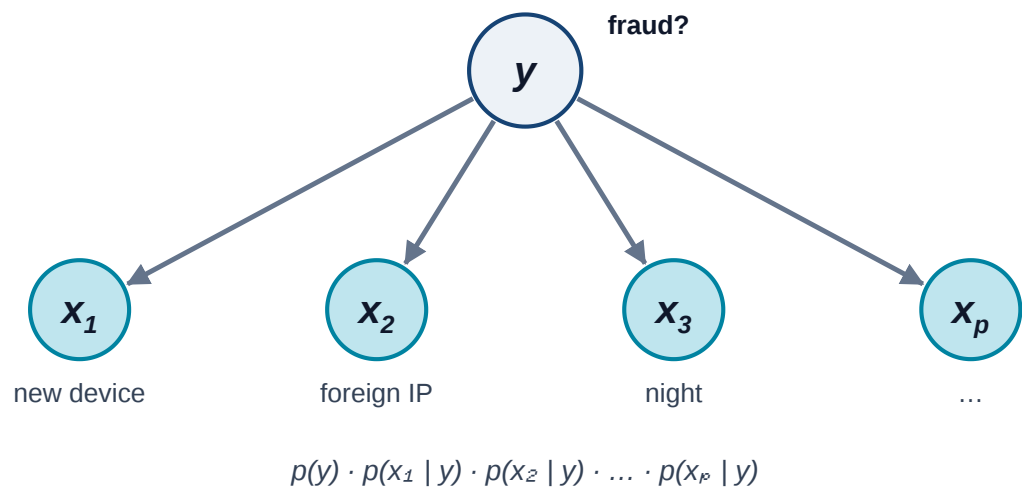
Speaker notes

Math first, then the reading (P5). This is the definition naive Bayes rests on, so it comes before naive Bayes (the old order taught it a module later). The full treatment — a counted example, and the case where conditioning *creates* dependence — opens Lecture 05, where it becomes the language of graphs.

Leave the question open. Most students will say "yes, account take-overs bring both" — which is exactly the dependence naive Bayes will assume away, and the subject of "Counting Evidence Twice" at the end of the module.



The Naive Assumption



Assume the attributes are **independent given the class**:

$$p(x_1, \dots, x_p \mid y) = \prod_{j=1}^p p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 3 — assume it apart. Dropped: every dependence *within* a class.

Speaker notes

The graph is a preview of Lecture 05, where a missing arrow will be read as an independence claim: here *no* arrow joins two attributes, so every pair is claimed independent given the class. Naive Bayes is the most extreme picture that is still a model.

The count: one Bernoulli per flag per class, p numbers instead of $2^p - 1$. The mini-exercise of Module 1 (15 numbers for three flags) becomes $2 \cdot 3 + 1 = 7$.

Name the shortcut as in the Lecture 01 table and say what it drops. The model still keeps $p(y)$ and $p(x | y)$, so it is still generative; it just refuses to see how attributes move together.



Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \quad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y)}{n(y)}$$

Lecture 02

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli.

Lecture 03

Its maximum likelihood estimate is the **relative frequency** — the same derivation as h/n for the coin.

Naive Bayes is fitted in one pass over the table, with no optimiser — and every number it uses is a **peak**. Shortcut 1 is taken as well.

Speaker notes

Math first: show the two formulas and ask what principle produces them before revealing the cards verbally. The answer the course has prepared: the categorical distribution was introduced in L02 as "the distribution naive Bayes uses for every nominal attribute", and L03 derived that the MLE of a Bernoulli is h/n . Under the naive assumption the likelihood splits into one categorical per attribute and class, so each is maximised separately — by counting.

This is worth a sentence: the independence assumption is what makes the MLE a closed-form count. Logistic regression has no closed form.



The Classifier

$$P(y \mid x_1, \dots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^p \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio LR}_j}$$

Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

The first line is Bayes with the naive product, and the denominator is total probability — nothing new, only assembled. The odds form is what makes the method readable: each flag is one factor, and the order in which they arrive does not matter. Odds were introduced with Bayes in L02.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds 1 : 999.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?



Speaker notes

Let the room guess first; most will say "clearly fraud". The model here has exactly these three flags, all of them present, so only the three ratios enter. With a flag observed *absent* its factor would be $(1 - 0.5)/(1 - 0.2) = 0.625$ for night, for example — worth mentioning if someone asks.

Numbers: $0.6/0.1 = 6$, $0.7/0.05 = 14$, $0.5/0.2 = 2.5$; $6 \cdot 14 \cdot 2.5 = 210$; $210/(210 + 999) = 0.1737$.



Three Red Flags

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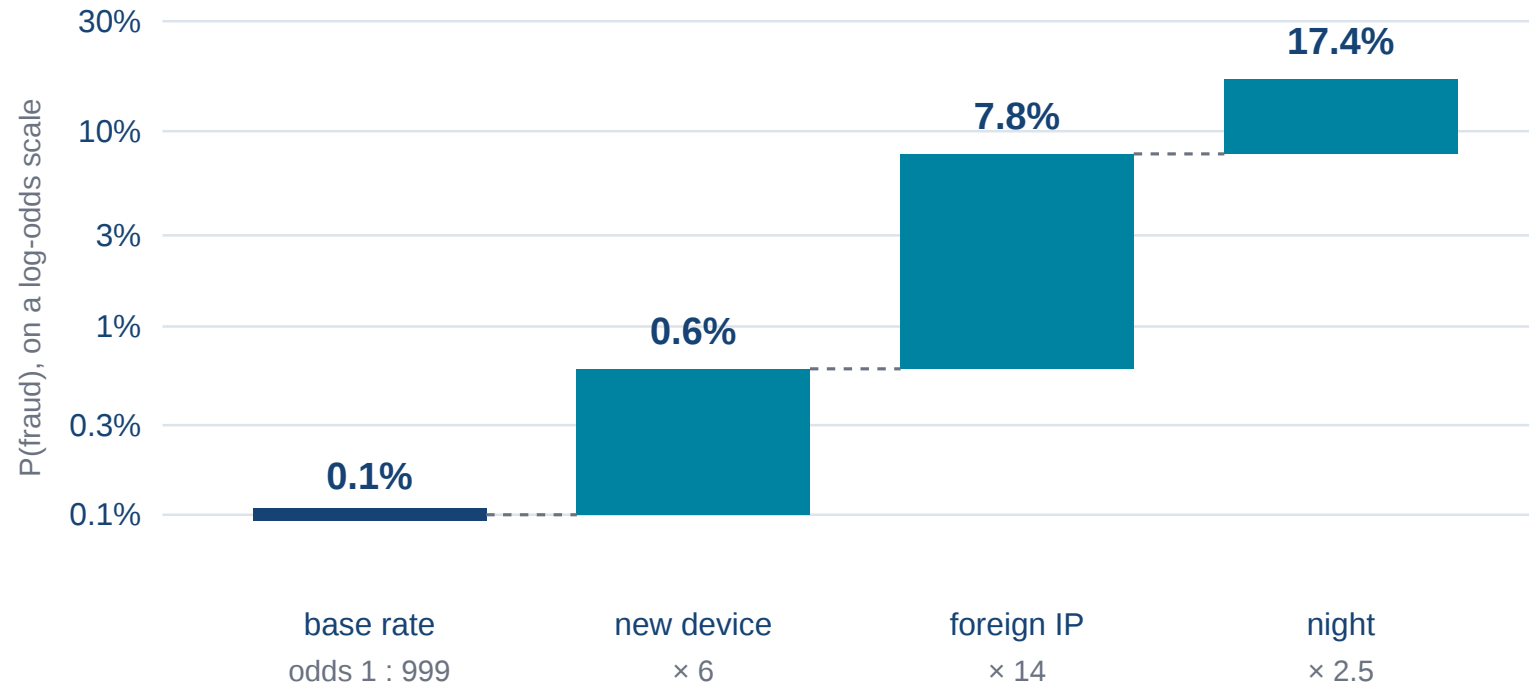
A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \implies P(\text{fraud} \mid x) = \frac{210}{1,209} \approx 17\%$$

Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up

odds 210 : 999 $\rightarrow$ $P(\text{fraud}) = 17.4\%$



Predict before each click: which flag moves the probability most?

Speaker notes

Stepped: one click per flag. On a log-odds axis every likelihood ratio is a step of fixed height, independent of where we start: $\log 6$, $\log 14$, $\log 2.5$. So the evidence *adds*. Stop after the foreign IP (7.8%) and ask what the night flag will do; students tend to overestimate it because "night" sounds suspicious, but 2.5 is the weakest ratio.

Plant the seed for Module 3: a sum of per-attribute contributions plus a starting value is a linear score — the form of logistic regression.



One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

KEEP THE PEAK, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — the three heads in three flips of Lecture 03, in a single cell.

Speaker notes

The zero-frequency problem, in the course's language: an MLE on a small count is extreme. With 100 frauds, many plausible fraud patterns have count zero. Ask for a repair before the next slide — "add one" usually comes up. The next slide shows that "add one" is not a hack.



Smoothing Is a Prior

Laplace smoothing adds one pseudo-count to every value of attribute j (k_j values):

$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$ — the **posterior mean** $\frac{a+h}{a+b+n}$ of Lecture 02 under the flat prior Beta(1, 1). For $k_j > 2$ values, the flat Dirichlet does the same.

count among 100 frauds	MLE (peak)	smoothed (mean)
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

Under the flat prior the peak of 0 of 100 is still 0. Smoothing quietly **leaves the peak for the mean** — exactly where the peak was wrong.

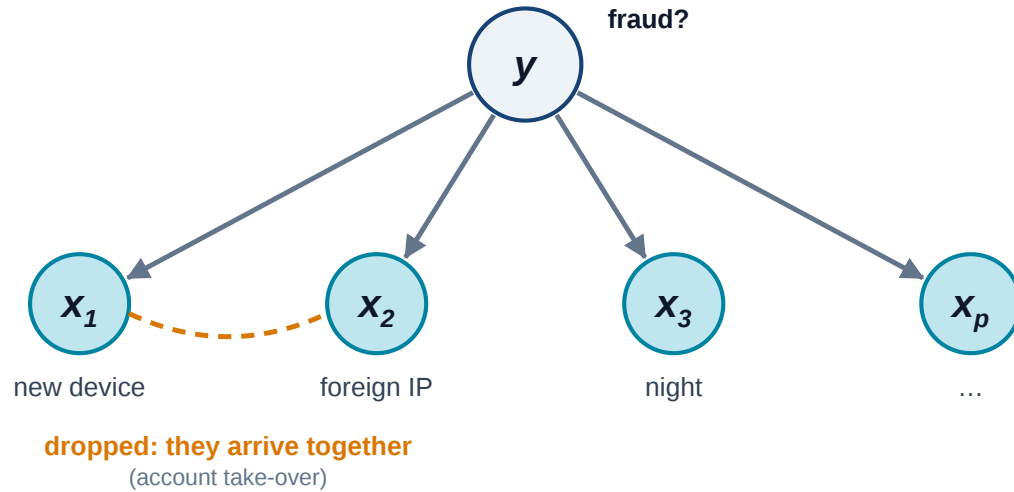
Speaker notes

This is the most satisfying tie of the lecture: a folklore trick of naive Bayes is the posterior mean the students computed in L02 (after 3 of 10 with a flat prior: mean $4/12 = 0.33$). The Dirichlet is the Beta for more than two values; it needs only the name here.

The table shows why nobody notices the switch on well-populated cells (0.60 vs 0.598) and why it matters on empty ones. Larger pseudo-counts mean a stronger prior, as in L02's "near a half" analyst.



Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Speaker notes

Answer the question left open on the first slide. The dropped edge is drawn dashed: naive Bayes multiplies both ratios as if they were separate witnesses, though they largely report the same event. With the pair's own ratio 13.75, the posterior odds are $13.75 \cdot 2.5/999 = 34.4/999$, i.e. 3.3% instead of 17.4%.

The consequence connects to Lecture 03's calibration: naive Bayes scores push towards 0 and 1 and are not calibrated. That argmax decisions often survive anyway is the classic explanation of why naive Bayes classifies well (Domingos & Pazzani, 1997) — the answer to the module's guiding question.



Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Speaker notes

One question, answerable qualitatively in seconds; the number is for checking ($2940/3939 = 0.746$). The bug is the extreme case of the previous slide: perfect correlation. Real feature sets contain near-copies all the time — the same information coded twice, derived features, one-hot groups.



Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Up — from 17% to about 75%. The odds get a second factor 14: 2,940 : 999. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and **keeps the peak** of every count; smoothed, it keeps the Beta posterior *mean* instead. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

Generative vs Discriminative

Guiding Question: Naive Bayes and logistic regression see the same fraud table. Which would you trust with 50 labelled frauds — and which with 50,000?

The Same Line, Fitted Twice

Take the logarithm of the naive-Bayes odds. For yes/no flags $x_j \in \{0, 1\}$ with $\theta_{jy} = P(x_j = 1 \mid y)$:

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = w_0 + \sum_{j=1}^p w_j x_j, \quad w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Where have you seen a log-odds that is linear in x ?

Math first (P5). The derivation is two lines and belongs on the exercise sheet: $\log p(x_j | y) = x_j \log \theta_{jy} + (1 - x_j) \log(1 - \theta_{jy})$, so each flag contributes a constant plus x_j times a weight; the constants and the log prior odds collect in w_0 . With the fraud numbers, w for the new device is $\log 6 - \log(0.4/0.9) \approx 1.79 + 0.81 = 2.60$.

The consequence of "jointly": logistic regression sees that the new device and the foreign IP arrive together and splits the credit between them. Naive Bayes cannot; each weight is estimated as if its flag were alone. That is the double counting of Module 2, now visible in the weights.

The Same Line, Fitted Twice

Take the logarithm of the naive-Bayes odds. For yes/no flags $x_j \in \{0, 1\}$ with $\theta_{jy} = P(x_j = 1 \mid y)$:

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = w_0 + \sum_{j=1}^p w_j x_j, \quad w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Where have you seen a log-odds that is linear in x ?

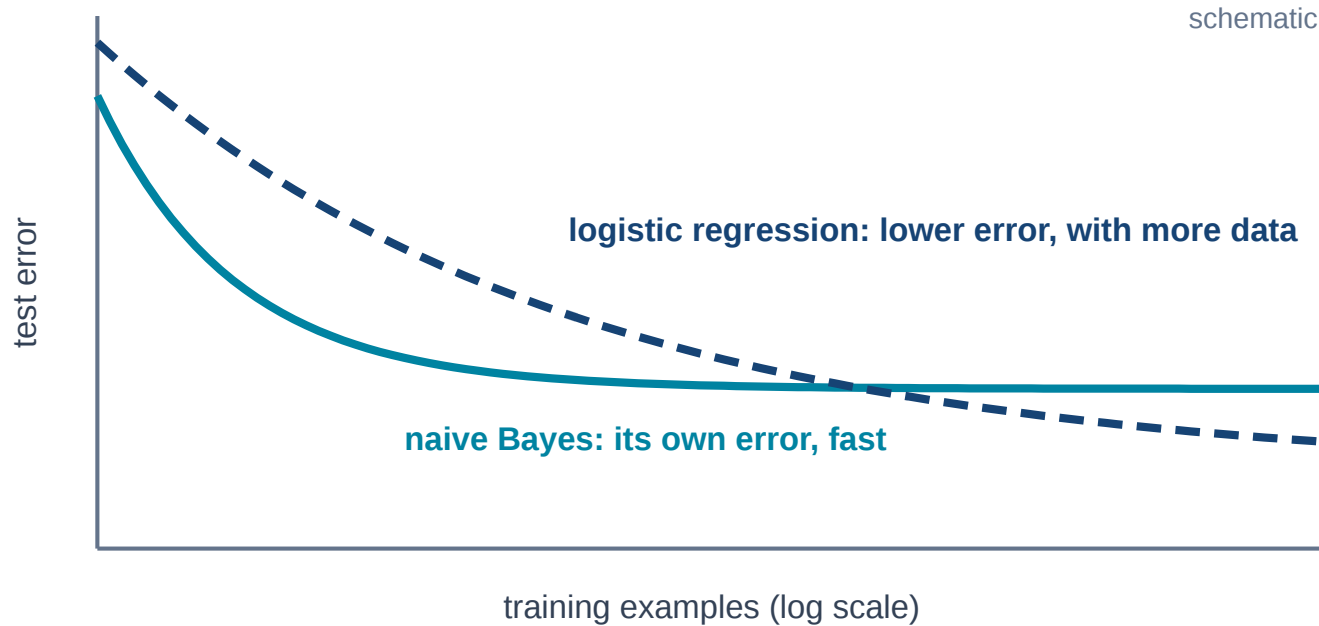
Logistic regression (Lecture 03): $P(\text{fraud} \mid x) = \sigma(w_0 + w^\top x)$. Same family of classifiers — naive Bayes sets the weights by **counting**, one attribute at a time; logistic regression by **maximising** $\sum_i \log p(y_i \mid x_i)$, all weights jointly.

Two Ways to a Classifier

	naive Bayes — generative	logistic regression — discriminative
models	$P(y)$ and $p(x y)$	$P(y x)$ only
fitted by	counting (joint likelihood)	iterative optimisation (conditional likelihood)
correlated flags	counted twice	credit shared
a missing field	drop its factor	needs a value
an unusual x	low $p(x)$: can be flagged	still a confident score
a new base rate	change $P(y)$	refit or correct the intercept
drops	dependences between attributes	everything about x

A table because the two are compared along shared dimensions. Walk it row by row and let students predict each cell for logistic regression before showing it, if time allows. The last row is the course arc: each model is placed by what it drops. "Correct the intercept": in logistic regression the base rate sits inside w_0 , so a new base rate shifts w_0 by the log of the ratio of prior odds — possible, but not visible in the model as a separate factor.

Few Data, Many Data



Few labelled frauds: naive Bayes. Its assumption acts like a strong prior — little to estimate, so it settles fast.

Many labelled frauds: logistic regression. It pays for fewer assumptions with data, and ends lower.

Shown empirically and in theory by Ng & Jordan (2002). A wrong assumption costs accuracy in the limit and buys it early — the prior of Lecture 02, again.

Speaker notes

The figure is schematic — say so. Ng & Jordan (NIPS 2002, "On discriminative vs. generative classifiers") show that naive Bayes approaches its asymptotic error with a number of examples logarithmic in the number of attributes, logistic regression with a number linear in it; the discriminative asymptote is never worse. On many real datasets the curves cross.

Answer the module's guiding question: with 50 labelled frauds, trust naive Bayes' ranking; with 50,000, logistic regression. And the general point: an assumption is information. When it is roughly right it substitutes for data; when it is wrong it caps what data can do.



Mini-Exercise

Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

The generative advantage in its most practical form. Marginalising a missing attribute out of a product of per-attribute factors is summing one factor over its values, which gives 1. The number: $15/1014 \approx 0.0148$. Compare with 17% when the IP was present and foreign: the missing field matters, and naive Bayes reports that honestly instead of guessing it.

Mini-Exercise

Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

Naive Bayes drops the factor: $\sum_v p(x_{\text{IP}} = v \mid y) = 1$, so the odds become $6 \cdot 2.5 : 999$ — about **1.5%**. **Logistic regression** has no model of x : it needs *some* value for x_{IP} , so someone must impute one, or train a second model without the field.

Takeaway

Naive Bayes and logistic regression draw **the same kind of line**; they differ in what they model. The generative model pays with assumptions about x and gets answers about x ; the discriminative one pays with data.

lecture	shortcut	what is dropped
03	keep the peak	how sure we are
03	only $p(y \mid x)$	every question about x
04	assume it apart	every dependence between attributes

Naive Bayes is a graph with one hub. **Lecture 05** draws the others — and keeps exactly the dependences we claim.

Close the lecture on the arc. The table is the L01 shortcut table, now with three rows filled. The teaser: the star graph of Module 2 is the simplest graphical model; Lecture 05 lets the modeller choose which arrows to keep, so the assumption becomes a statement rather than a blanket.

